

Research Methods

In Psychological Science

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1. RESEARCH METHODS

Why do we need research methods?

- Way of solving certain problems: when simple observation or common sense not enough, or when the outcome is uncertain
- Avoid errors in human reasoning: The human brain simplifies decision-making by using shortcuts ('heuristics') or ignoring information
- Avoid biases due to self-serving arguments or prejudice: Wanting to be right or push one's idea

Numbers are facts that can't be argued with

- Statistics are used to describe groups of numbers
- Research methods are a technique we use so that we can be confident about the results

number facts number facts number fc
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1. RESEARCH METHODS

What are the goals of research?

Describe

- Define an event or behaviour
- Differentiate between two events
- Identify relationships

Understand

- How or why events occur
- What affects or influences the event or behaviour

Predict

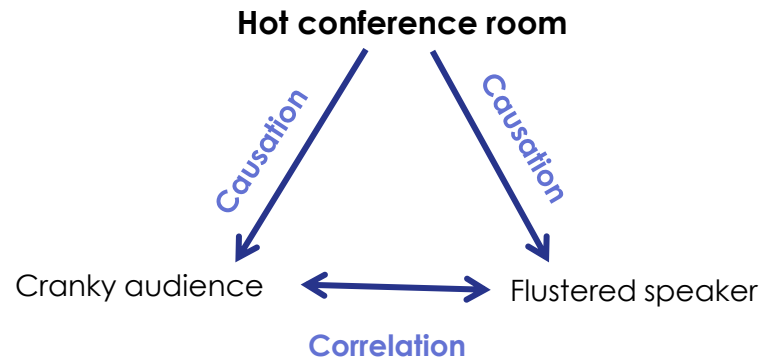
- Predicting behaviours, and understanding causes

.....human actions, thoughts and behavior

1. RESEARCH METHODS

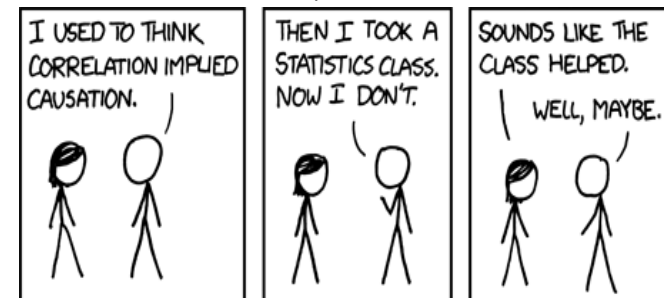
Note:

Correlation is not the same as causation



Causation =
* No other possible explanation for the effect (if the variable was taken out, could the effect still occur?)

Source: <https://xkcd.com/552/>



1. RESEARCH METHODS

What are the characteristics of research ?

- Rigorous methods are the foundation for all statistical analyses & inferences
- Fixed process: Question -> gathering data -> analyses -> reporting findings
- Open to scrutiny: Process includes making the results available to others (safeguards against bias or bad decisions)

2. THE RESEARCH PROCESS



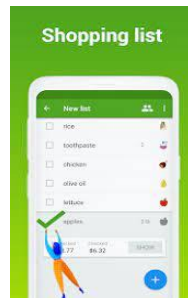
1



Ask question

Formulate a question and a research hypothesis

2



Planning

Plan your study

3



Collection

Collect data

4



Prepare (analyse)

Analyse the data

5



Display results

Display the data

6



Interpret

Interpret the data – does it fit the theory (hypothesis)

3. STARTING WITH A RESEARCH QUESTION



Determining the objective (purpose) of the research

- Prior notions or observation, or want to develop an idea
- Want to find an answer to a practical problem or issue
- Begin with an assumption



Developing the research question and hypothesis

- Grand theories are too general/broad and difficult to test
- Best to have firm, specific, practical research questions
- Narrow down to a specific and practical issue (makes planning, data collection and interpretation easier)



A research hypothesis

- What you intend to test in the study
- Makes a prediction about the results (based on previous results)
- Statistical analyses will test the null hypothesis - **H₀** – (nil effect)

3. STARTING WITH A RESEARCH QUESTION

Scientific vs. non-scientific questions (Not all questions require statistical investigations!)

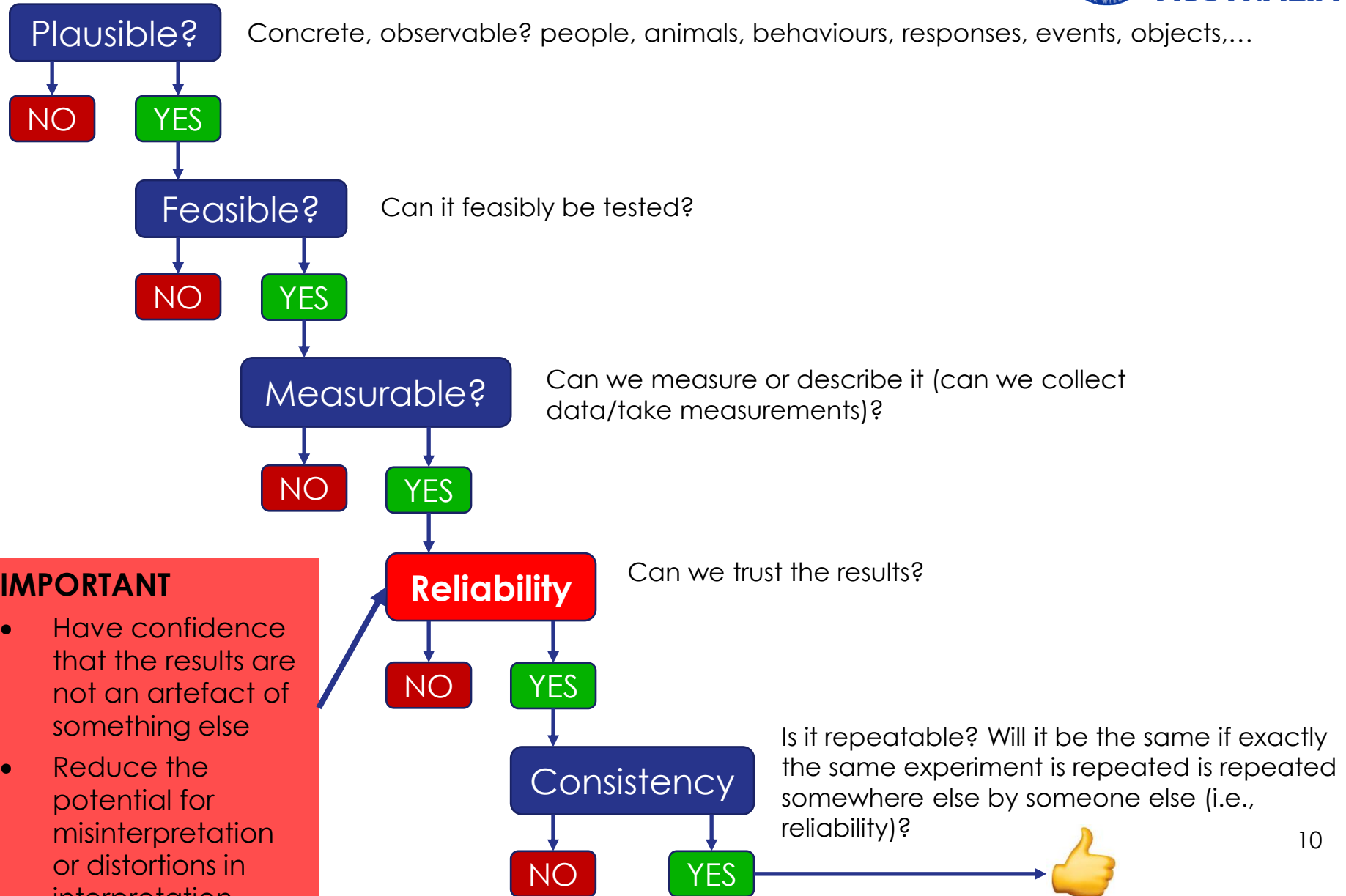
Unscientific Questions

- Vague, or too large/complex (World peace, poverty)
- Unobservable, can't be accurately measured, or not based on science (ghosts, religion)
- Different results for different people on different occasions

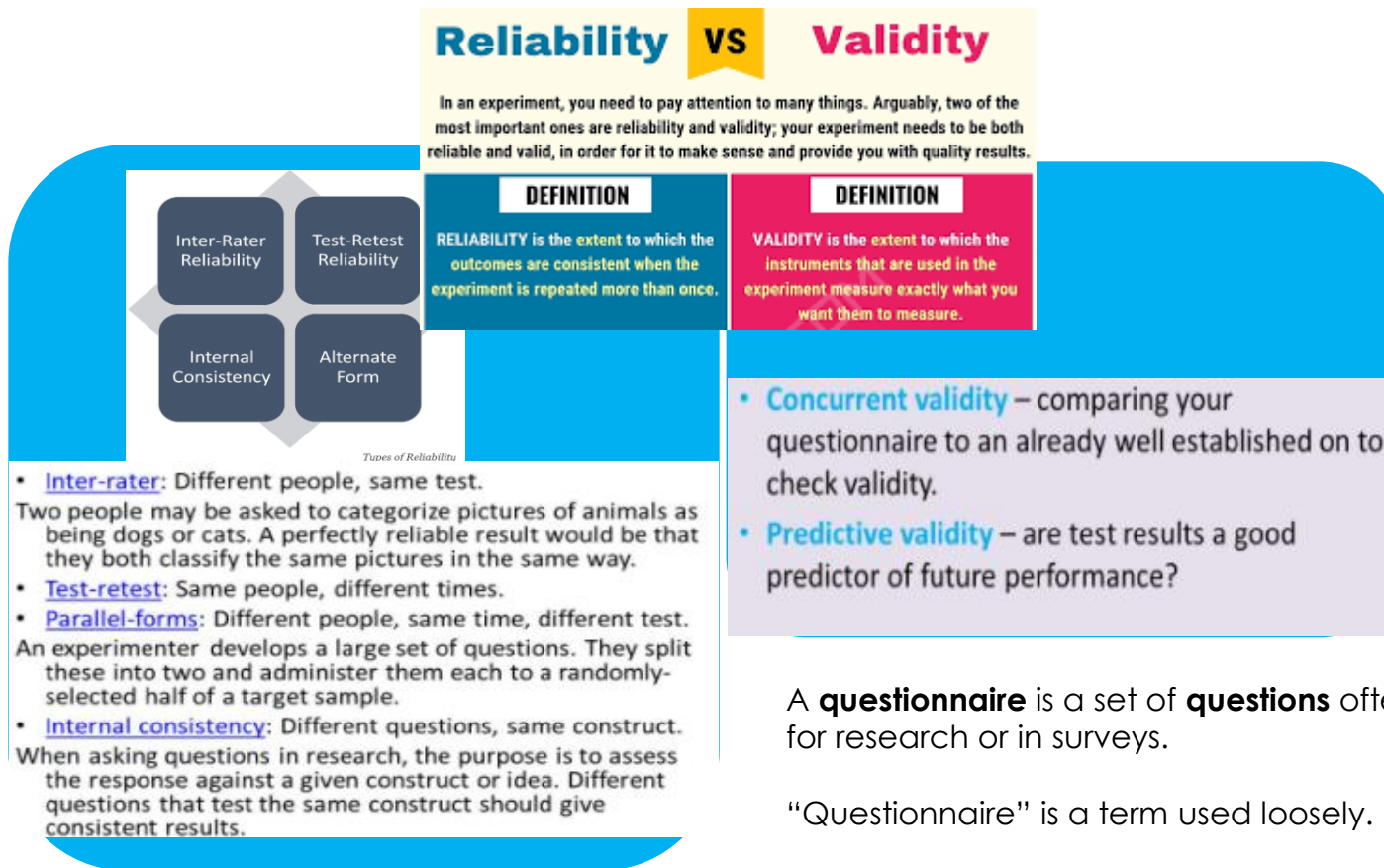
Scientific Questions

- Can only be answered after data collected and analysed
- Often involves a large number of people or repeated measurement
- Make inferences about connections between multiple events or factors
- Statistics allow us to make definite and valid conclusions about data

Scientific vs non-scientific questions



When designing a new questionnaires (scale, measure)



A **questionnaire** is a set of **questions** often used for research or in surveys.

“Questionnaire” is a term used loosely.

It can refer to a list of questions put together by an experimenter for a survey.

A good questionnaire has been extensively developed and tested for reliability and validity.

- Framing a research question: PIOC (or PICO) acronym

P	I	O	C
P opulation	I ssue (Independent variable)	O utcome measure (dependent variable)	C omparison, C ontrol, C onfounding variable



Framing a research question: PIOC (or PICO) acronym

P 	I 	O 	C 
Population	Issue (Independent variable)	Outcome measure (dependent variable)	Comparison, Control, Confounding variable
<i>In...</i>	<i>Does.....affect</i>	<i>....this outcome</i>	<i>..compared to ...taking into consideration..</i>
Year 11 and 12 students	Time spent on social media	Maths exam results	Teacher A or B
High-school students	Vaping (frequency/amount)	Memory - Self-report (1 = very bad, 10 = very good)	When started vaping
Older people	Vaccine x	Blood clots	Younger people
Young drivers (L and P plates, aged 17-18)	Vehicle speed	Car accidents	Distraction, alcohol

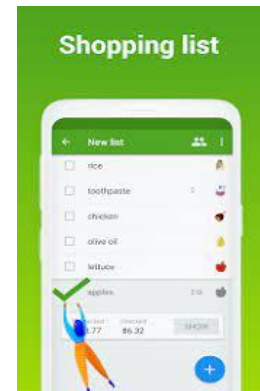
4. PLANNING THE STUDY

Thinking ahead

Once the research question has been identified, the methodological issues need to be addressed.

Good planning helps to:

- Provide evidence that is strong enough to support/deny the assumptions
- Have confidence that the results are not an artefact of something else
- Reduce the potential for misinterpretation or distortions in interpretation
- Have as much control (reduce influence of co-variables) over the results
- Reduce variability or biases in the data to be collected



4. PLANNING THE STUDY

How to get good data



Consider:

- **WHO:** The participant sample. Who should be included/excluded. Using mixed method, there may be different participant samples (e.g., high school students, teachers, parents)
- **WHAT:** What are the main concepts/variables to be examined? There needs to be a close relationship between the research question and the variables. What are the independent and dependent variables? How will you measure them?
- **WHEN/WHERE:** When is the best time to collect the data? Where should it take place (e.g., school ground, outside shopping centre)?
- **HOW:** Method or technique of information gathering (e.g., survey, questionnaire, experiment)

4. PLANNING THE STUDY.... With a template

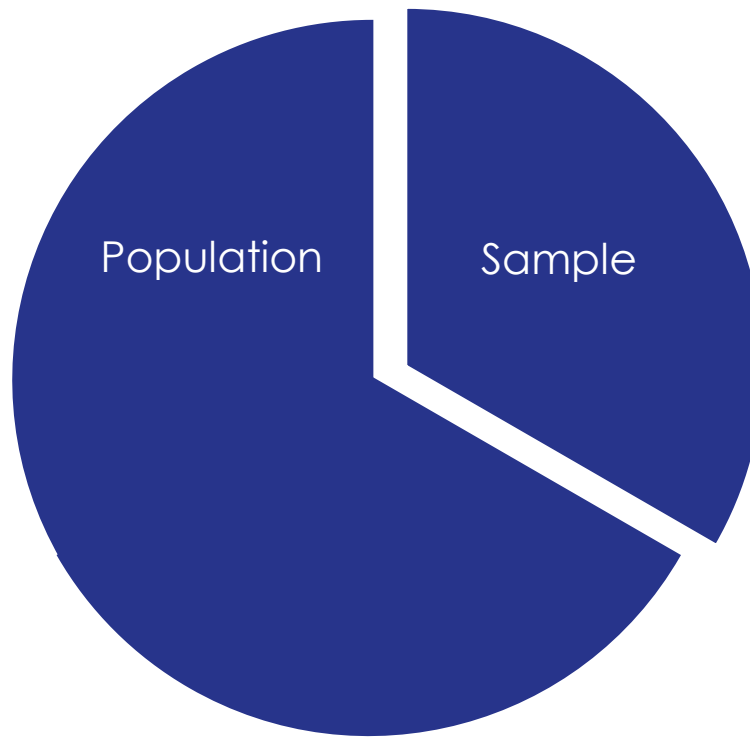
TEMPLATE				
PROJECT TITLE:				
RESEARCH QUESTION:				
WHO (participants, sample)		WHAT (variables)	WHEN OR WHERE	HOW (Methods)
Inclusion	Excluded (not target group, or confounders)	(to address inclusion and exclusion criteria)		
E.G. PROJECT TITLE: <i>Marijuana and memory in high school (HS) students</i> RESEARCH QUESTION: <i>Do high school students who use marijuana have worse self-reported memory than high school students who have never used marijuana?</i>				
WHO (participants, sample)		WHAT (variables)	WHEN OR WHERE	HOW (Methods)
Inclusion	Excluded (not target group, or confounders)	(to address inclusion and exclusion criteria)		
HS students, 2 groups: * Group 1: Usage > 1 cone/joint/hit per week for 3+ months * Group 2: never used	Not HS students * Group 1: Infrequent use (< 1 per week) * Group 1 or 2: has used other drugs at any time	- drugs use (frequency, duration, amount) - memory - age, gender, education level + Other questions that may be prove or disprove the question	- Schools in the same SES - Recess times or just before/after school	Surveys, Questionnaires Qualitative interviews
TEMPLATE				
PROJECT TITLE:				

4. PLANNING THE STUDY



Sampling

- We can't test everyone so we need to choose
- Sampling = choosing a group that is small enough to study, but which it still **representative** of the population being studied.
- The sample size is important as it affects the amount of variability in the sample (think: pick 2 people at random – how representative of pop?)
- Fundamental to all statistics:
 - Is the basis for determining whether the difference between groups is statistically different
 - Can make inferences about the results that the difference is not likely to be due to chance



Population



Sample

Ideally, the sample will be representative of the broader population (then the results can be generalised to the population).

However, the Sample might have a list of inclusion and exclusion criteria (*e.g. aged 16-18, from School Y and Z only*), so this will limit the generalisability

4. PLANNING THE STUDY



Sampling techniques

- The aim is to reduce the risk of researcher bias, and to give everyone in a population an equal chance of being selected
- A system is used to select who to approach for research

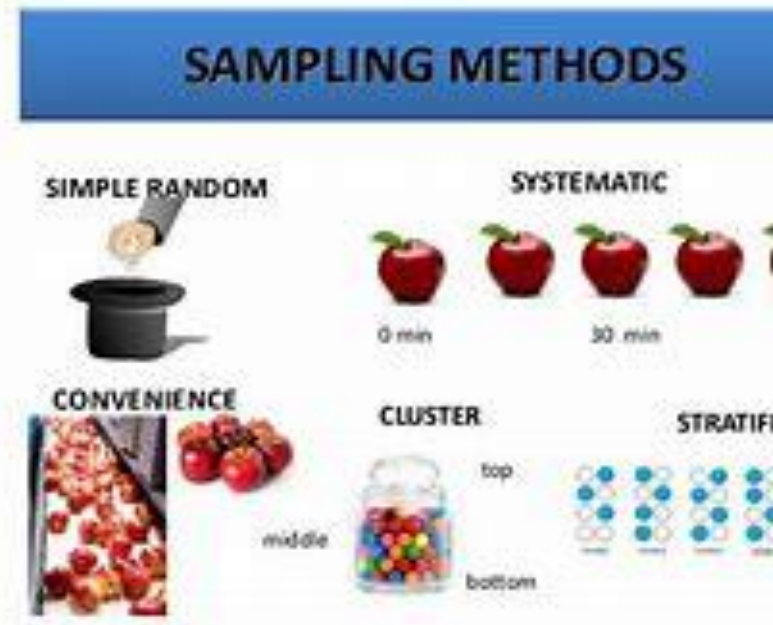
Convenience sampling: Recruited easily (convenience, low cost)

Most rigorous is to have a **sampling frame** (list which covers everyone in the population), and then select a subsample:

Simple (random) sampling: Pick randomly or use random number generator

Systematic sampling: Start on the list at a random point, then pick at fixed intervals (e.g., 20th)

Stratified sampling: The aim is to reduce the random sampling errors by selecting participants based on subgroups ('stratifying') (e.g., genders, age groups)



Type of errors

TYPE OF ERRORS	Problem	Solution
Sampling errors	The sample is not representative of the whole population. Could be due to: • incorrect coverage of the target population, • incorrect sampling frame, • Selection bias from the experimenter • Non-response (or bias) from participants • Random due to chance variation	
Non-sampling (or systematic errors)	Errors or flaws in how the research design was designed or, or conducted. May be due to 1. <u>Design problems</u>: errors when designing the study 2. <u>Measurement</u> errors: Weaknesses in study design or measurement, or the questionnaires or technique is not appropriate for the methods, or issues in data scoring 3. <u>Researcher errors</u>: Errors when interpreting the results (inferential errors); Might only select people most likely to support the hypothesis 4. Random bad luck	

- Systematic error in sampling, design, data collection or analysis will affect estimate of true effect

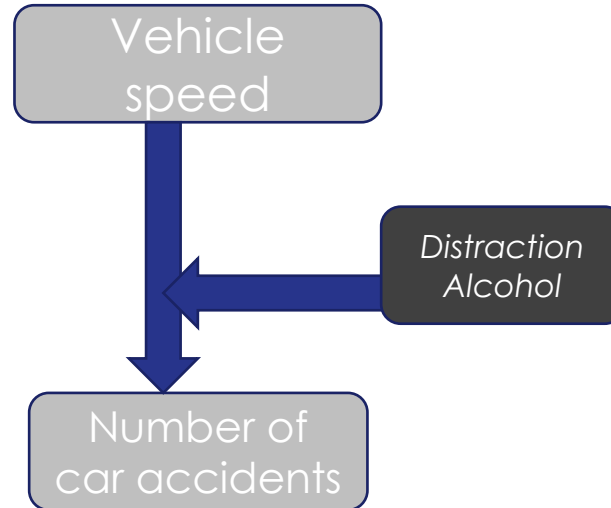
4. PLANNING THE STUDY



Independent variable (IV):

Is the antecedent condition that occurs prior to the dependent variable.
Must have more than 2 values or groups (it must be variable).

Is often manipulated (controlled) by the experimenter (e.g., different groups, treatment condition, before and after therapy)



Confounding variable:

Uncontrolled variable that can affect or distort the nature of the relationship between the IV and DV.

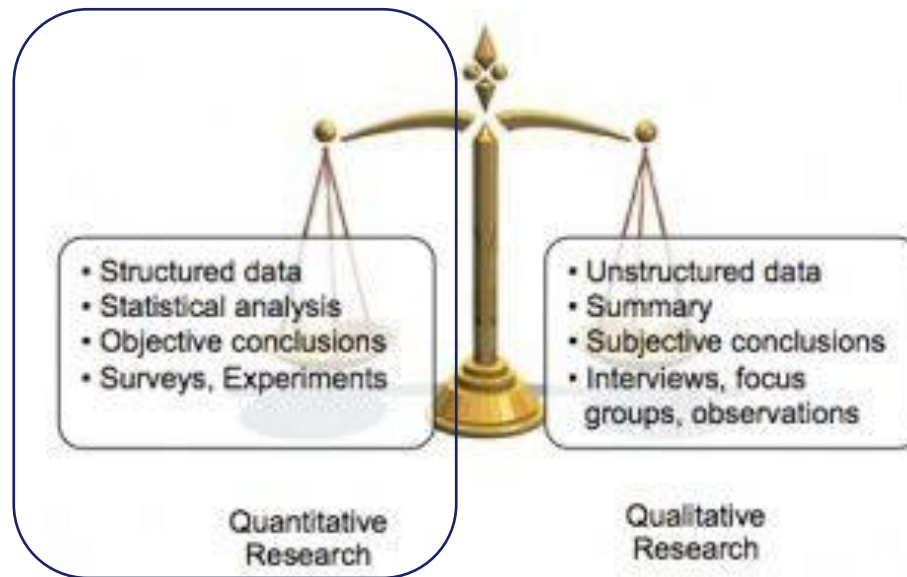
Car accidents cannot only be attributed to vehicle speed.



Outcome: Dependent variable (DV)

Is observed in order to assess the effects of treatment.

Typically is a measurement score. The change in the score *depends* on the independent variable.



Quantitative research	Tests specific hypotheses Uses rigorous scientific methodologies and measurements Aims to provide an explanation that is supported using statistical methods	Data is • Entered in tables • Summarised using descriptive statistics • Reported using numbers • Statistically explained	<u>Advantages:</u> • Can include large samples • Most rigorous methodology <u>Drawbacks:</u> • Narrow focus can miss broader picture or real life elements, often makes small incremental steps in knowledge	Observational - Descriptive Correlational Experimental (comparative) Quasi-experimental
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QUANTITATIVE study designs

	Description	Advantages/disadvantages
Observation Descriptive	Describes the characterises events or group of individuals, and the frequency with which some events occur, and their relationship	<u>Advantages:</u> Can reveal new issues or trends and be used to generate new hypotheses <u>Disadvantages:</u> Not as scientifically valid because of the lack of control over all the variables
Experimental	Tests the interaction or cause/effect relationship between events. Often are the only designs that can test causality! Researcher is firmly in control of the research conditions and the research participants - All conditions are tightly controlled to isolate the effects of specific variables. - This helps to rule out alternative explanations (due to the confounding effects of factors other than the independent variable) * Randomised Control Trials use Randomisation: Researchers assign participants to different conditions (often treatment) as it will reduce the impact of bias from factors that are not measured	<u>Advantages:</u> Gold standard in research evidence Most rigorous All conditions are tightly controlled, so can be repeated exactly at another location with different participants <u>Disadvantages:</u> Representative of 'lab' situations (i.e., less representative of the real world)
Quasi- experimental	Similar in content to experimental, but subjects are not randomised Less powerful than experimental research	<u>Advantages:</u> Easier to conduct (less restrictions on sampling); More reflective of real world <u>Disadvantages:</u> Cannot control all variables (risk that unknown variables affected the results)
Survey	A list of questions addressing only the issues of interest Surveys produce numerical descriptions of the social world (e.g., 20% of cats prefer a brand) Designed to compare cases systematically, and search for relationships and links between variables	<u>Advantages:</u> Can be conducted using face-to-face, telephone or online; Can provide quick answers in large sample <u>Disadvantages:</u> Does not control all variables. It relies on large numbers to overcome population variability, and low response rate will affect the generalisability of results

Before data collection

5. Human research ethics principles, and informed consent



National Statement on Ethical Conduct in Human Research

(NHMRC 2007), Australian Code for the responsible conduct of research (2018)

Principles	What it means
Research merit	• Importance (impact) of the project (Is the project worthwhile, does it make a significant contribution to knowledge or well-being?) • Sound methodological design (the methodology can answer the research question)
Research integrity	• Appropriate Research Skills • Qualified persons/team • Research conduct (Honesty, lack of bias, no conflict of interest) • Record keeping (privacy and data security) • Allow scrutiny of findings
Justice	• Recruitment without discrimination (Fair selection of participants) • No unfair burden • No exploitation • Benefit benefits to, participants or to society
Beneficence	• Balance of risk and benefits • Risks must not exceed the benefits
Respect	• <u>Privacy</u>: information that is collected is protected (only collect what you need, stored with passwords and held securely) • <u>Confidentiality</u>: Not disclosing information to other people* *(NOTE THE LIMITS "UNLESS REQUESTED BY LAW OR DANGER TO SELF-OTHERS") Informed consent: • Enough information to understand what it involves (purpose and methods), what are the risks or adverse effects (including distress) • Privacy and confidentiality statements (how, where and how long data will be stored) (limits of confidentiality) • Respecting their decision • Autonomy to say no without consequences • Respect to change their mind at any time

Informed consent in vulnerable participants

	Population	Considerations
Capacity to give consent is impaired or compromised	• <u>Mental Illness</u> • <u>Drugs and alcohol</u> Capacity to provide informed consent may change over time.	Need to assess capacity to understand and remember what the project involves and the risks, and to make important decisions
	• <u>Learning or intellectual impairment</u> • <u>Neurodegenerative disease (e.g. dementia)</u> Capacity to provide informed consent will reduce over time	If capacity is compromised, a carer or guardian needs to provide informed consent after consultation with medical practitioner (Guardianship Act, 2021 – best interest, no objection)
Legal capacity	<u>Children</u>: A minor is someone under the age of 18.	Young children: ethics requirements: • Explain the procedures and what it involves in simple language • Must agree to take part ('assent') • Parents/guardian must provide informed consent 'Mature minors' • Ethically difficult area for ethics (generally > 15 y/o) • Are able to make informed decisions and to understand what the project involves
Other vulnerable population groups	• Aboriginal and Torres Straight Islanders • Pregnant women	Have their own ethics committee, review process and values to inform research activities

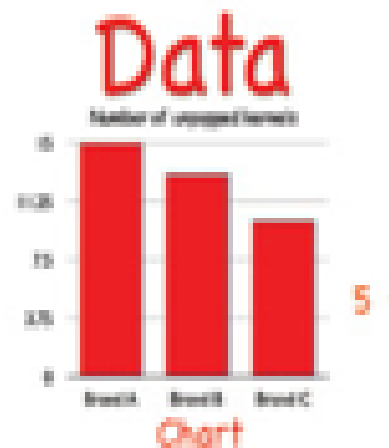
Informed consent in young people about a sexual act/behaviour



Ensuring informed consent from the other person

1. Are they legally capable to make a decision?	Do they have the capacity to make an informed decision? • Legal capacity: Check aged 16+ • Mental capacity: not under the influence of drugs/alcohol, learning disability, mental health issues
2. Do they have the freedom to say yes or no (i.e., voluntary)?	They are completely free to say yes or no (no pressure, no coercion)
3. Are they fully informed?	about what is going to happen (no deception) ... and the consequences (e.g., are they likely to become distressed?)
4. They have the right to withdrawal of consent	Consent is not permanent and can be withdrawn at any time without any consequence

6. Data collection, preparation and analysis



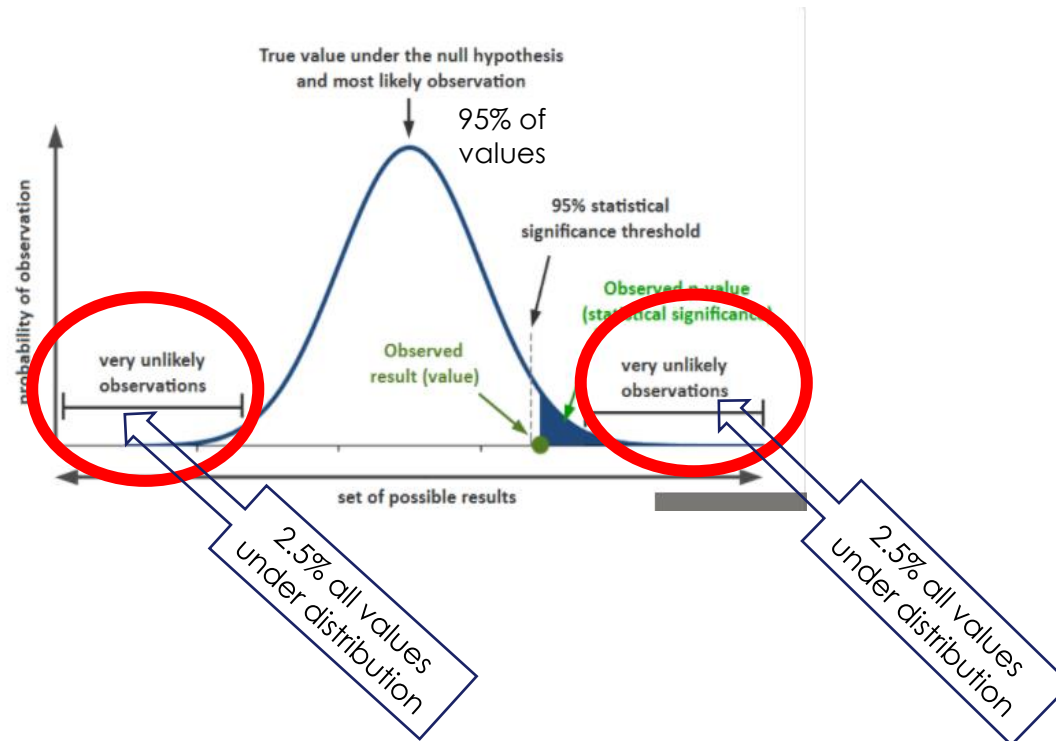
HYPOTHESIS TESTING

Hypothesis testing uses sample data to evaluate a hypothesis about a population

Statistical way to determine whether the result occurred by chance alone.

Is based on the area under the curve in each tail of the standard normal distribution below:

P-value significance level (threshold) is $0.05 = 5\%$ (Statistical threshold used for deciding whether the results are due to chance or unlikely.

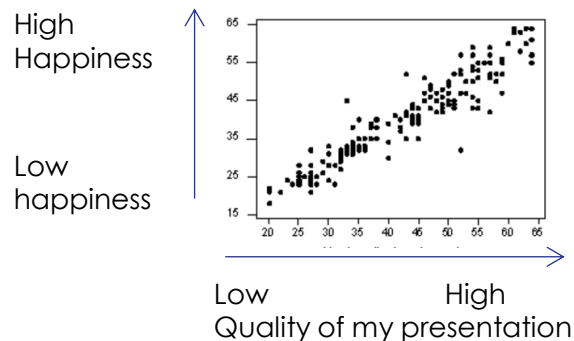


HYPOTHESIS TESTING

Example: Is there an association between happiness (measured from 0-10), and the quality of my presentation (measured 0-10)?

We run a Pearson test on our statistical software, and it shows the following result (Pearson rho = 0.93), with significant level **p = 0.025**

- **P = 0.025** is the same as 2.5%. So, there is a **2.55%** chance that the association between your happiness and my talk is random (or 97.45% not due to chance).
- Because $p = 0.02$ is less than 0.05, we conclude there is a significant association.
- We look at our scatter graph to observe the direction of the results: People who enjoyed my talk were also very happy.



(note it is an association not causation, as happy people could rate presentations higher).

HYPOTHESIS TESTING

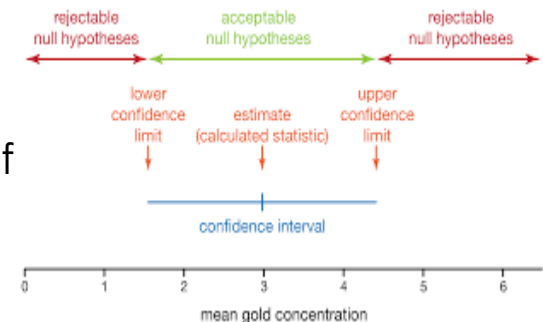
YIPPEEE!

$p < 0.05$: STATISTICALLY SIGNIFICANT

- $0.05 = 5\%$
- So $p < 0.05$ = less than 5% chance that our results unlikely, random or due to chance

$P < 0.05\%$ meets threshold for deciding yes or no.

- 100% (all results) – $5\% = 95\%$
- We are 95% confident that the effects are meaningful
= there is a significant association between X and Y

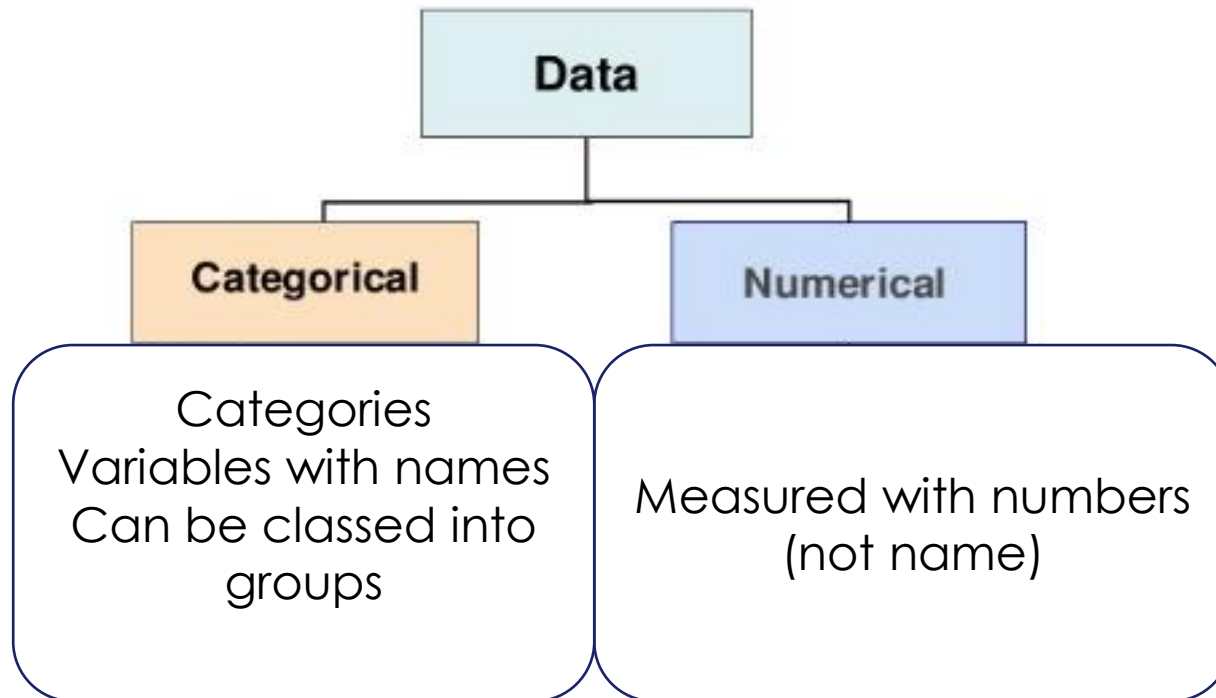


$p = 0.05$ or $p > 0.05$: NOT STATISTICALLY SIGNIFICANT

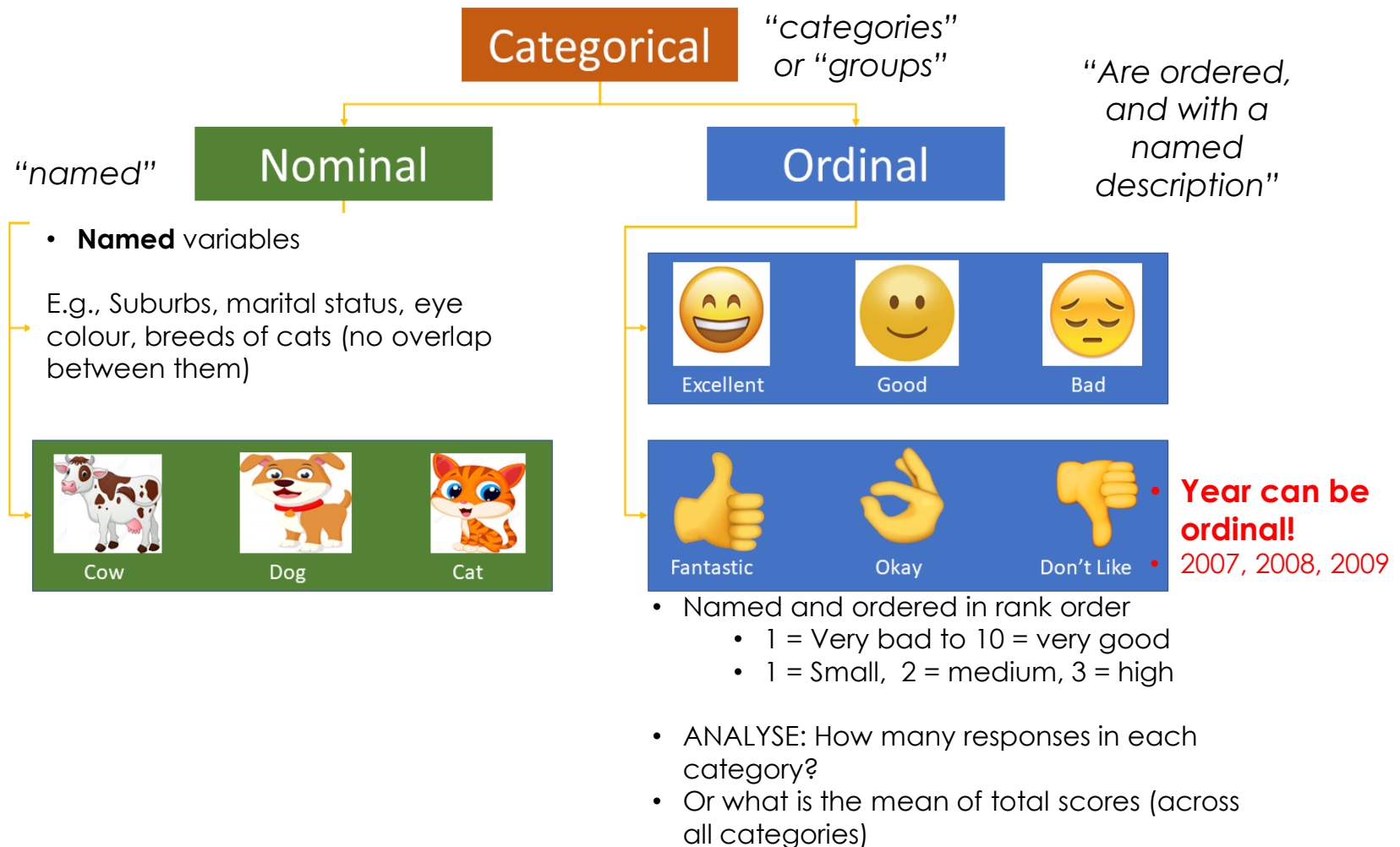
- Example: $p = 0.10$, which means 10% chance that the results are unlikely, random or due to chance
- We are not fully confident that the results reflect a true effect (we are only 90% confidence)

DATA types

- May involve simple or complex statistical techniques
- Data types dictate how you summarise and visualise the results



DATA types: Categorical data



DATA types:

Turning numerical into categorical

Age	Tally (head count)	Frequency count (number of people in this group)	Percentage
15		19	25.7% $(19/74)*100$
16		10	13.5%
17		30	40.5%
18		15	20.3%
Total		74	100%

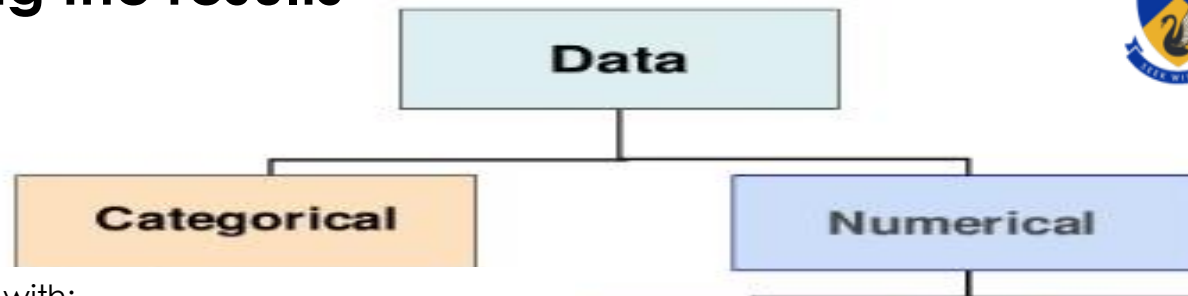
Table 1: Summary of results (using age as a numerical variable)

	Mean	Min	Max
Age	16.5	15	18

Table 2: Summary of results (turning age into categories)

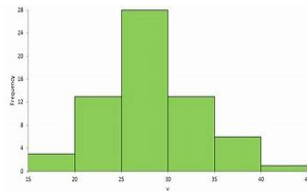
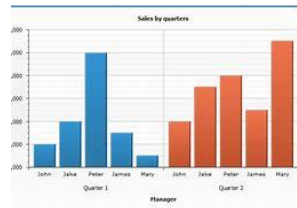
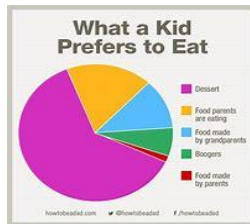
	Frequency count	Percentage
Age		
15-16	29	39.2%
17-18	45	60.8%

7. Displaying the results



Summarise with:

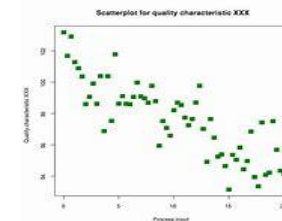
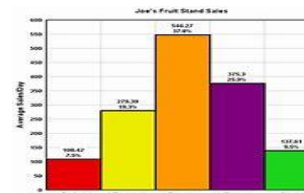
- Counting responses in each category (**frequency** or 'count')
- Proportions:** count of categories divided by total number of observations
- Percentage:** Proportion as decimal x 100%
- Pie chart, bar graphs, histograms



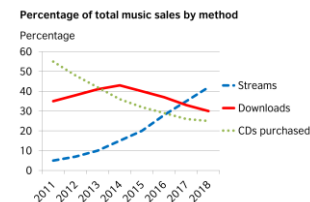
Histogram: often displays frequencies (counts)
Will show the count observations in each groups or category (e.g. number of children, in 2018, 2019, or number of single and married people)

Summarise with:

- Average ('mean')
- Median, mode, min, max, standard deviation
- Display with bar graph (Mean or median happiness rating in audience A and B)
- Scatter graph: uses dots to represent all values across two different axes (one dot is one person)

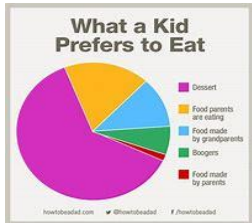


- Line graph: changes change in values (mean, or %) over time

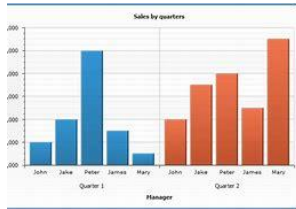


7. DATA visualisation – what do you want to do?

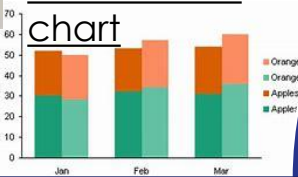
Pie chart
(few categories)



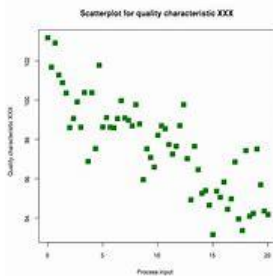
Bar chart



Stacked bar chart



Scatter plot



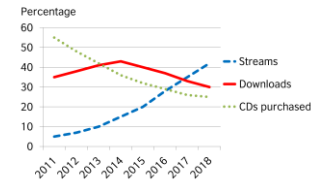
Compare & contrast categories

Show change over time

Show relationship between 2 variables

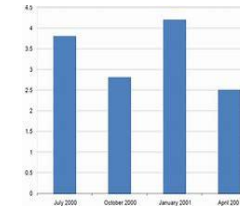
Show distribution or display data

Percentage of total music sales by method

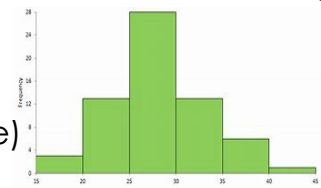


Line graph
(many categories)

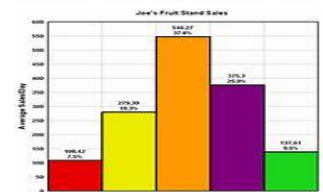
Bar graph
(fewer categories)



Histogram
(count of.., one variable)



Bar graph
(frequency count, percentages or averages)





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